HEC-COMPLIANT CURRICULUM

8-Week Python & AI/ML Internship Blueprint

A Detailed Operational Plan & Resource Guide for Pakistani University Students

Target Audience: BS CS/SE/DS/AI Students **Duration:** 8 Weeks (2 Months) **Focus:** Industry-Ready Deployment

1. Introduction & Pedagogical Strategy

This document provides a highly detailed framework designed to bridge the gap between academic theory taught across Pakistani universities and the rigorous standards expected by both local and international technology ecosystems. Aligned with the Higher Education Commission (HEC) internship directives, this plan ensures students spend 30–40 hours per week working through structural objectives, building reproducible code, and deploying real-world solutions. It prioritizes open-access, high-quality documentation and targeted dataset discovery over generic search practices.

2. Weekly Blueprint: What to Look For & Where to Look

Week 1: Advanced Python, OOP, and Modern Workflows

🎯 **WHAT TO LOOK FOR**

- **Environment Isolation:** Correct architecture using `venv` or `conda` to separate dependencies.
- **OOP Clean Architecture:** Encapsulation, inheritance, and clean Magic Methods (`__init__`, `__str__`, `__repr__`).
- **Git Practices:** Branch management, atomic commits, descriptive messages, and resolving merge conflicts.

🔍 **WHERE TO LOOK**

- **Official Python Docs:** The *Python Tutorial* sections on Errors/Exceptions and Classes.
- **Real Python:** Comprehensive guides on "Python Virtual Environments: A Primer" and "Object-Oriented Programming (OOP) in Python 3".
- **GitHub Skills:** The official interactive labs by GitHub for learning branching and conflict resolution.

Week 2: The Data Science Stack (NumPy & Pandas)

🎯 WHAT TO LOOK FOR

- **Vectorization vs Loops:** Utilizing NumPy universal functions (`ufuncs`) instead of slow `for` loops.
- **Pandas Idioms:** High-performance index alignment, boolean indexing, `.loc`/`.iloc`, and handling structural missing data (`NaN`).
- **Statistical EDA:** Identifying outliers, multi-collinearity via correlation matrices, and distribution shapes.

🔍 WHERE TO LOOK

- **Pandas Official Documentation:** The "10 Minutes to pandas" guide and the API Reference on Essential Basic Functionality.
- **Kaggle Datasets:** Locate localized datasets such as *Pakistan Real Estate Data (Zameen.com scraps)* or *Pakistan E-commerce Dataset*.
- **Seaborn Gallery:** Pre-built structural layouts for heatmaps, pairplots, and violin distributions.

Week 3: Supervised Learning Foundations

🎯 WHAT TO LOOK FOR

- **Data Splits & Leakage:** Ensuring strict validation isolation (Train/Test split).
- **Mathematical Optimization:** Understanding cost functions like Ordinary Least Squares (OLS) and Cross-Entropy loss.
- **Evaluation Metrics:** Moving past basic accuracy to track Precision, Recall, F1-Score, and R^2.

🔍 WHERE TO LOOK

- **Scikit-Learn User Guide:** Section 1.1 (Linear Models) and Section 1.10 (Decision Trees).
- **UCI Machine Learning Repository:** Look up foundational tabular datasets (e.g., Housing, Wine Quality).
- **StatQuest with Josh Starmer:** YouTube video breakdowns for Linear Regression, Logistic Regression, and ROC/AUC curves.

Week 4: Ensemble Methods & Hyperparameter Optimization

🎯 WHAT TO LOOK FOR

- **Bias-Variance Tradeoff:** How bagging reduces variance and boosting lowers bias.
- **Search Spaces:** Setting up parameters for `GridSearchCV` and `RandomizedSearchCV`.
- **Early Stopping:** Preventing overtraining in gradient-boosted trees.

🔍 WHERE TO LOOK

- **XGBoost/LightGBM Docs:** Official documentation covering parameters tuning and python API references.
- **Machine Learning Mastery:** Detailed tutorials by Jason Brownlee on ensemble methods and tuning architectures.
- **Kaggle Competitions:** Reviewing winning solutions and notebooks utilizing ensemble frameworks.

Week 5: Unsupervised Learning & Dimension Reduction

🎯 WHAT TO LOOK FOR

- **Clustering Validation:** Finding optimal clusters using the Elbow Method and Silhouette Analysis.
- **Variance Retention:** Tracking Explained Variance Ratio in Principal Component Analysis (PCA).
- **Feature Scales:** The absolute necessity of standardizing data before calculating distance metrics.

🔍 WHERE TO LOOK

- **Scikit-Learn Clustering:** Section 2.3 (Clustering methods documentation and parameter breakdowns).
- **Towards Data Science:** Practical code applications for PCA visual walk-throughs and customer clustering.
- **DataCamp Open Community:** Tutorials focused on cluster profiles and interpreting components.

Week 6: Introduction to Deep Learning & Generative AI APIs

🎯 **WHAT TO LOOK FOR**

- **Neural Topography:** Layers, activation functions (ReLU, Softmax), and gradient descent backpropagation.
- **API Mechanics:** Managing API keys securely using environment variables (`.env` files).
- **Prompt Structuring:** System prompts, user messages, and controlling temperature configurations.

🔍 **WHERE TO LOOK**

- **Hugging Face Course:** Chapters covering basic Transformer pipelines and fine-tuning introductory models.
- **OpenAI** / **Google AI Documentation:** Official Quickstart guides for API requests and JSON schema validation.
- **DeepLearning.AI:** Free short courses covering prompt engineering and basic LLM application strategies.

Week 7: System Integration & Model Deployment (MLOps)

🎯 **WHAT TO LOOK FOR**

- **Model Persistence:** Serializing weights using `joblib` or `pickle` safely.
- **API Architecture:** Designing clean REST endpoints using FastAPI with input schemas validated by Pydantic.
- **State Management:** Keeping UI responsive during inference rendering in Streamlit.

🔍 **WHERE TO LOOK**

- **FastAPI Tutorial User Guide:** Step-by-step documentation on path parameters, query parameters, and request body structures.
- **Streamlit API Reference:** Detailed layouts on session states, data caching, and metric displays.
- **Docker Documentation:** Multi-stage build patterns for small, efficient Python containerization.

Week 8: Documentation, Viva Preparation, and Compliance

🎯 **WHAT TO LOOK FOR**

- **Code Transparency:** High-quality docstrings (Google or NumPy standard) and comprehensive README layouts.
- **Technical Defense:** Preparing answers for model failure modes, bias, and performance tradeoffs.
- **Compliance Check:** Ensuring all log books are filled out weekly and signed off.

🔍 **WHERE TO LOOK**

- **HEC Pakistan Portal:** Reviewing the explicit framework for university industrial internship compliance guidelines.
- **Make a README Guide:** Structural examples of world-class open-source project documentation layouts.
- **GitHub Gists:** Standard technical interview sheets for ML engineering and core data frameworks.

3. HEC-Compliant Evaluation Matrix

To qualify for academic credit matching university systems, performance parameters are strictly segmented across the timeline:

Component	Weighting	Audit Vector	Frequency
Weekly Progress Logs	20%	Verifiable commits on GitHub; clear markdown entries matching weekly deliverables.	Weekly
Mid-Term Assessment	20%	Live coding check examining basic Pandas wrangling and baseline Scikit-Learn training.	End of Week 4
Capstone Execution	40%	Code health, error mitigation, container functionality, and UI reactivity.	End of Week 7
Technical Defense (Viva)	20%	Oral examination, evaluation of trade-offs made, and full project presentation.	End of Week 8

💡 **Academic Advisory:** Students are strongly encouraged to select localized Capstone domains (e.g., Urdu processing, domestic financial trend forecasting, or regional climate anomalies) to standout within competitive job markets.